SOLUTION

SECTION–I

1. Ans. (B)

Sol. $f'(a) = f'(b)$

$3a^2 + a = 3b^2 + a \quad \Rightarrow a = -b$

so $f(1) = 1 + a + b = 1$

2. Ans. (C)

Sol. $f'(x) = 3(x^2 - 2ax + a^2 - 1)$

Clearly roots of $f'(x) = 0$ must be distinct and lie in the interval $(-2, 4)$.

$\therefore \quad D > 0 \quad \Rightarrow \quad a \in R \qquad .....(1)$

$f'(-2) > 0 \quad \Rightarrow \quad a^2 + 4a + 3 > 0$

$\Rightarrow \quad a < -3 \text{ or } a > -1 \qquad .....(2)$

$f'(4) > 0 \Rightarrow \quad a^2 - 8a + 15 > 0$

$\Rightarrow \quad a > 5 \text{ or } a < 3 \qquad .....(3)$

and $-2 < -\frac{B}{2A} < 4 \Rightarrow -2 < a < 4 \qquad .....(4)$

From (1), (2), (3) & (4), $a \in (-1, 3)$

3. Ans. B

Sol. Using LMVT in [2, 4]

$$f'(c) = \frac{f(4) - f(2)}{4 - 2} = \frac{f(4) + 4}{2}$$

$$f'(x) \geq 6 \Rightarrow \frac{f(4) + 4}{2} \geq 6 \Rightarrow f(4) \geq 8$$

4. Ans. (C)

Sol. Let $x = 3\sqrt{3}\cos\theta$, $y = 3\sqrt{3}\sin\theta$ where $\theta \in [-\pi, \pi]$

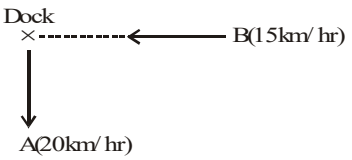
Then $A = x - y = 3\sqrt{3}(\cos\theta - \sin\theta)$

Now $-\sqrt{2} \leq \cos\theta - \sin\theta \leq \sqrt{2}$

$\therefore \quad A_{max} = 3\sqrt{3}\sqrt{2} = \sqrt{54}$

5. Ans. (A)

Sol.

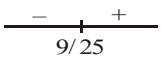


At 2 PM boat B is 15 km away from Dock. At time t hr boat A is

20t km way & boat B is 15 – 15t km away from dock.

$(AB)^2 = f(t) = 400t^2 + 15^2(1 - t)^2$

$f'(t) = 800t + 15^2 \cdot 2(t - 1) = 1250t - 450$



AB is minimum at $t = \frac{9}{25}$ hrs. = 21 min 36 sec.

$\therefore \quad$ time = 2:21:36 PM

6. Ans. D

Sol. $f(-2) = f(3) = 0$

f(x) is continuous in [–2, 3] & derivable in (–2, 3) so Rolle's theorem is applicable.

so $\exists \ c \in (-2, 3)$ such that $f'(c) = 0$

$$\Rightarrow \quad \frac{2c^3 - 5c^2 + 4c - 1}{(c-1)^2} = 0 \quad \Rightarrow \quad c = 1/2$$

7. Ans.(C)

Sol. $y = \frac{1}{x}$

so $x^2 + y^2 = x^2 + \frac{1}{x^2}$

minimum value = 2

8. Ans. (C,D)

Sol. $f(x) = 3x^2 + e^{-x}\int_0^x e^t f(t)dt$

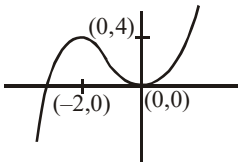
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$$f'(x) = 6x - e^{-x}\left(\frac{f(x)-3x^2}{e^{-x}}\right) + e^{-x}e^{x}f(x)$$

$f'(x) = 3x^2 + 6x$

$f(x) = x^3 + 3x^2$ $\quad [\because \quad f(0) = 0]$



9. Ans. (B,C)

Sol. By LMVT for $f(x)$ in $x \in [1, 2]$

$\exists\ c_1 \in (1, 2)$ where

$$f'(c_1) = \frac{f(2)-f(1)}{2-1}$$

$\because \quad f'(x) \geq 1\ \forall\ x \in R$

$\Rightarrow \quad f(2) - f(1) \geq 1 \quad \Rightarrow \quad f(2) \geq 2$

By LMVT for $f(x)$ in [2, 3]

$\exists\ c_2 \in (2, 3)$ where

$$f'(c_2) = \frac{f(3)-f(2)}{3-2}$$

$\Rightarrow \quad 4 - f(2) \geq 1 \Rightarrow f(2) \leq 3 \Rightarrow 2 \leq f(2) \leq 3$

10. Ans. (B,C,D)

Sol. (B) $F(x) = f(3-4x) \Rightarrow F'(x) = f'(3-4x)(-4)$

$\Rightarrow \quad F'(x) < 0$

(C) $G(x) = f(x^2 - x)$

$\Rightarrow G'(x) = f'(x^2 - x)(2x - 1)$

$G'(x) > 0 \Rightarrow 2x - 1 > 0$

(D) $H(x) = (f(x))^3 \Rightarrow H'(x) = 3(f(x))^2 f'(x)$

Paragraph for Question 11 to 12

$$f(x) = x + 2\sqrt{25 + (\ell - x)^2} \quad \text{..... (i)}$$

$0 \leq x \leq \ell$

11. Ans. (B)

Sol. $f'(x) = 1 - \dfrac{2(\ell - x)}{\sqrt{25 + (\ell - x)^2}}$ .....(ii)

$$= \frac{\sqrt{25 + (\ell - x)^2} - 2(\ell - x)}{\sqrt{25 + (\ell - x)^2}}$$

$25 + (\ell - x)^2 = 4(\ell - x)^2$

$\sqrt{3}(\ell - x) = \pm 5$

$x = \ell \pm \dfrac{5}{\sqrt{3}}$

$x = \ell - \dfrac{5}{\sqrt{3}}$

If $\quad \ell = \dfrac{185}{\sqrt{3}}$ $\quad$ then $\quad x = \dfrac{180}{\sqrt{3}} = 60\sqrt{3}$ $\quad$ kms.

12. Ans. (D)

Sol. $\ell = 10$ kms, then maximum value of $f(x)$ lies at $x = 0$ or $x = 10$

$f(x) = x + 2\sqrt{25 + (10 - x)^2}$

$f(0) = 10\sqrt{5}$

$f(10) = 20$

$\Rightarrow$ maximum freight $= 10\sqrt{5}$ at $x = 0$

Paragraph for Question 13 & 14

13. Ans. (B)

Sol. $f(x) = 2\cos^2 2x + \dfrac{3}{4}\sin 4x + ax$

$f'(x) = -4\sin 4x + 3\cos 4x + a \geq 0\ \forall\ x \in R$

$a \geq 4\sin 4x - 3\cos 4x$

$a \geq 5$

14. Ans. (D)

Sol. $f'(x) \neq 0$ for any $x \in R$

$a \notin 4\sin 4x - 3\cos 4x$ for any $x \in R$

$\Rightarrow \quad a \notin [-5, 5]$

$a \in (-\infty, -5) \cup (5, \infty)$

SECTION–III

1. Ans. 4

Sol. Let a point on $y^3 = x^4$ be (t^3, t^4)

$3y^2y' = 4x^3$

$\Rightarrow \quad y' = \dfrac{4x^3}{3y^2} \quad \Rightarrow \quad y' = \dfrac{4}{3}t$

Equation of tangent is

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$$y - t^4 = \frac{4t}{3}(x - t^3)$$

$\because$ it is a normal to $x^2 + y^2 - 2x = 0$

$\therefore$ it must pass through (1, 0)

$$\Rightarrow \quad -\frac{3}{4}t^3 = 1 - t^3 \quad \Rightarrow \quad \frac{t^3}{4} = 1$$

$$\Rightarrow \quad t^3 = 4$$

Now $m = \frac{4t}{3}$

$$\Rightarrow \quad \left(\frac{3m}{4}\right)^3 = t^3 = 4$$

2. Ans. 1

Sol. $R = \frac{abc}{4\Delta}$

$$= \frac{r.r.\left(2r\sin\frac{\theta}{2}\right)}{4\left(\frac{1}{2}r^2\sin\theta\right)} = \frac{r\sin\frac{\theta}{2}}{\sin\theta} = \frac{r}{2\cos\frac{\theta}{2}} = \frac{r}{2\cos 15^\circ}$$

$$\therefore \quad \frac{dR}{dr} \times 2\cos 15^\circ = 1$$

3. Ans. 9

Sol. Let $\frac{dV}{dt} = kV^2$

$$\Rightarrow \quad \int \frac{dV}{V^2} = \int k\,dt \quad \Rightarrow \quad -\frac{1}{V} = kt + C$$

at t = 0, $V = V_0$

$$\Rightarrow \quad C = -\frac{1}{V_0} \Rightarrow \quad \frac{1}{v} = \frac{1}{V_0} - kt$$

at t = 6, $V = 2V_0$

$$\Rightarrow \quad 6k = \frac{1}{2V_0} \qquad \Rightarrow \quad k = \frac{1}{12V_0}$$

when $V = 4V_0$

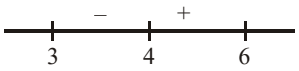
$$\Rightarrow \quad kt = \frac{3}{4V_0} \quad \Rightarrow \quad t = \frac{3 \times 12V_0}{4V_0 \times 1}$$

t = 9 hours

4. Ans. 1

Sol. $f(x) = \int_3^x t\left(t^2 - 7t + 12\right)dt$,

$f'(x) = x(x-3)(x-4)$

$\therefore$ 

minimum at x = 4

maximum at x = 3 or 6.

$$\therefore \quad f(3) = 0, \; f(6) = \int_3^6 (t^3 - 7t^2 - 12t)dt$$

$$= \left(\frac{t^4}{4} - \frac{7t^3}{3} + \frac{12t^2}{2}\right)_3^6 = \frac{99}{4}$$

$$f(x) = \int_3^4 (t^3 - 7t^2 + 12t)dt = -\frac{7}{12}$$

$$\therefore \quad 3a + \frac{b}{9} = 1$$

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